

MACHINE LEARNING PROJECT

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INTRODUCTION

The classification topic is a widely discussed theme in machine learning and statistics and is used in many different applications. The process consists in adapting a model on labeled data and the aim is to exploit the information given by the features in the dataset in order to correctly predict the labels of unseen instances. For this case study, the dataset chosen is the Saturn_Dataset.RDS created by professor Antonio Candelieri for the Machine Learning course. It contains 540 instances, described by 8 numeric features and a class label (that assumes values in $\{-1, 1\}$). Descriptive analysis of the dataset showed that feature 8 has greater variability compared to the others, that the data points are almost evenly split between the two true classes (275 obs with label -1 and 265 with $+1$), and that the features have low correlation with each other, except for features 1 and 7, and to a lesser extent, 1 and 3.

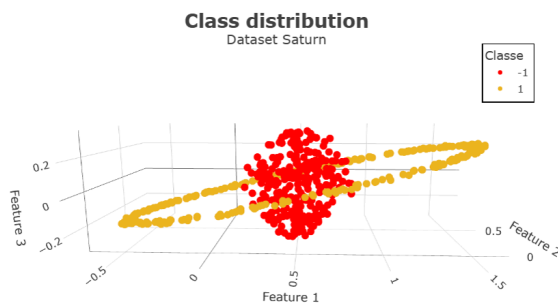


Figure 1.

METHODS

There exist several methods to perform classification. Among these, the ones that best

fit the specific dataset were chosen. The first valid approach is K-Nearest Neighbours (KNN), which assigns each new data point to the class that appears most frequently among its k nearest neighbours. It being a non parametric approach, leads to some advantages like the adaptability to various contexts and the absence of strong hypotheses. Furthermore, it's easily interpretable due to its simple structure. The only pivotal thing KNN requires is the hyperparameters selection: k (number of neighbours) and the type of distance.

The second method taken into account is Random Forest (RF). This algorithm creates a series of trees on several different bootstrapped versions of the original dataset (samples) using a fixed number of randomly chosen features for each split in the tree. This double randomization reduces the correlation between the trees which decreases variance and ensures a good capability in generalization. It can handle noisy, mixed and high dimensional data effectively. Like KNN, it is a non parametric model. Therefore, it requires the specification of some hyperparameters, the most important being the number of trees and the number of variables considered at each split.

Eventually, the last one is the Soft Margin Support Vector Machine (SVM). It divides the data through a hyperplane that maximizes the distance between the two margins while minimizing the classification error. The points that lie on or between the margins are known as Support Vectors (SVs) and are the core of the algorithm. The Soft Margin SVM implies the use

of C , a regularization hyperparameter for the empirical errors. SVM can be applied to non linear scenarios using a kernel function that maps data from an input to a feature space where the instances become linearly separable.

RESULTS

The following section analyzes the practical applications of the previously presented methods on the Saturn dataset. Indeed, starting with KNN, 10% of the dataset is left aside to test the performances of the final model and 10-Fold Cross Validation (CV) is applied to the remaining 90%. To optimally select the hyperparameter k , the mean of the empirical and generalization error are evaluated through the 10-Fold CV, for a series of k values between 10 and 40. A value of k is chosen arbitrarily among those that have a low difference between the empirical and generalization error, while ensuring that the generalization error does not increase by more than 10%.

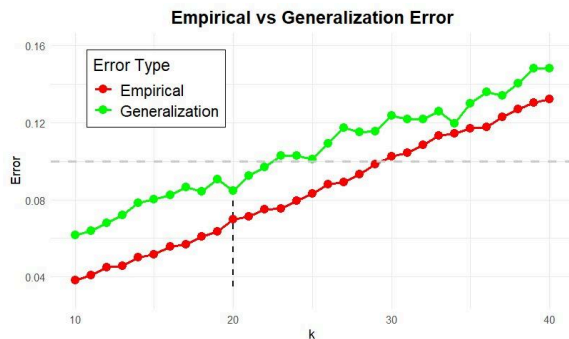


Figure 2.

As shown in Figure 2, with k equal to 20 and fixed a seed, both the empirical and generalization error are quite low and the prediction accuracy on the test set is 98.15%. Another seed, so consequently, another choice of k , could affect the results. However, based on the empirical observation, the k would likely be within a range of 20 to 25.

Going on with Random Forest, results are obtained employing 500 trees and 2 variables per split. As illustrated in Figure 3, the Out Of Bag error converges after about 100 trees and is almost zero. The shape of the two classes determines a difference in terms of error. A larger number of trees usually leads to a more stable model with better generalization.

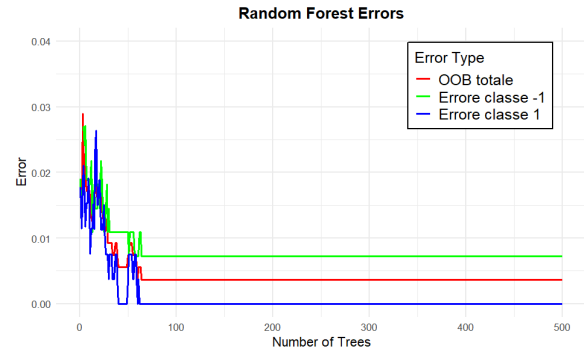


Figure 3.

To conclude, the last results obtained are the ones produced by Soft Margin SVM. At the start the model was trained without a kernel function, assuming that in some dimension the data could be separated linearly. This assumption is clearly not appropriate for the Saturn dataset. Consequently, considering the data distribution, the Radial Basis Function (RBF) kernel represents a more suitable choice. In addition to the regularization hyperparameter C , this type of kernel requires another hyperparameter γ (gamma) which determines how close two data points must be for the model to consider them similar. In order to estimate the two hyperparameters of the model, Bayesian Optimization (BO) is employed due to its computational efficiency.

In fact, as illustrated in Figure 4, the BO algorithm reaches the maximum accuracy (100%) faster than the Grid Search, with both methods considering identical ranges for each hyperparameter.

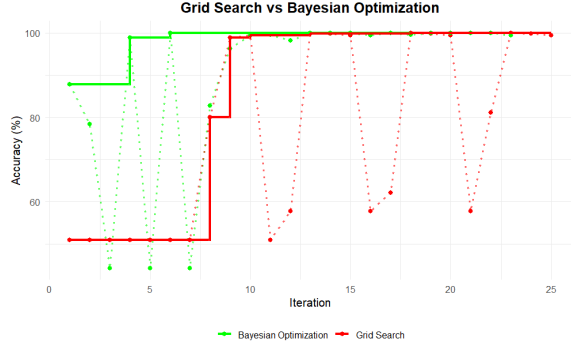


Figure 4.

The values chosen for the hyperparameters with the Bayesian Optimization are C equals to 31.45 and γ equals to 0.042. Given that the BO maximizing accuracy has nothing left to optimize once the 100% accuracy is quickly reached, it is used an alternative measure created to penalize models with an excessive number of SVs. With 12.14% of the total data points as Support Vectors, the model achieves a reasonable trade-off between the risk of underfitting or overfitting. In fact a low number of Support Vectors typically indicates that the model learns a clear decision boundary and possesses strong generalization capabilities.

CONCLUSIONS

In light of the above outcomes, the data seem to be divided into well separated classes. Thanks to that, a lot of models could effectively perform the classification task. The three methods discussed provide optimal results with few errors and, consequently, an almost perfect accuracy. Among all of them, the SVM model is preferred to KNN due to its best prediction accuracy and to Random Forest for its better interpretability. The Random Forest, indeed, works excellently with complicated and high dimensional data structures. However the current dataset is relatively simple and low dimensional, making such complexity unnecessary. On the other hand, the KNN is less

computationally efficient and more sensible to individual data points. The SVM reports a perfect accuracy and a great adaptability to the Saturn dataset. Overall, the results confirm that the SVM with RBF kernel represents the most suitable approach for this kind of non linearly separable data. Its ability to construct smooth and flexible decision boundaries, combined with strong generalization properties and robustness to noise, make it the most effective and reliable model among those tested.